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### **Task 1-Show skeleton tracking**

GitHub Link Of Demo : <https://github.com/Hasinireddy2407/AI-ML-INTERNSHIP>

- Skeleton lines connect body landmarks together to show the full body structure and proportions, helping AI understand where each body part is located and how they relate to each other.
- They make it easy to track body movement in real time across video frames because the connected lines move together as a unit making motion detection more stable and accurate.
- Skeleton lines convert the human body into a simple mathematical structure of points and connections that any AI model can understand regardless of the person's appearance, size or background.

### **Task 2 -Body Measurement Logic**

- **Shoulder Width Calculation**

Shoulder width is calculated by finding the Euclidean distance between the left and right shoulder landmarks detected by MediaPipe. This is one of the most important measurements in garment fitting research.

$$\text{Shoulder Width} = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

Where, (x1, y1) = left shoulder landmark coordinates

(x2, y2) = right shoulder landmark coordinates

Key points from research:

- MediaPipe returns normalized coordinates between 0 and 1 so they must be multiplied by image width and height to get pixel values
- Pixel distance is then converted to centimeters using a scale factor:  $\text{Real Width} = \text{Pixel Distance} \times (\text{Known Reference} / \text{Pixel Reference})$
- Shoulder width along with chest and hip measurements form the three primary parameters for upper body garment sizing according to anthropometric studies
- Average male shoulder width is 40-50 cm and female is 35-43 cm as per standard body measurement databases

### **Height Estimation**

Height estimation works by summing the distances between consecutive body landmarks from head to toe rather than taking a direct straight line distance because the human body is never perfectly straight in an image.

$\text{Height} = \sum \text{distance}(\text{landmark\_i}, \text{landmark\_i+1})$

Key points from research:

- The head to body ratio in an average adult is approximately 1:7.5 meaning if head length is known, full height can be estimated.
- Research papers use the nose landmark as the topmost point and ankle landmark as the bottommost point.
- A scaling factor is applied to convert pixel height to real world height in centimeters.
- Accuracy improves significantly when the person stands straight facing the camera at a known distance.

### **Waist Estimation**

Waist estimation is more complex because MediaPipe does not have a direct waist landmark. Research solves this by estimating the waist region from hip landmarks.

$\text{Waist\_Midpoint} = ((x_{\text{left\_hip}} + x_{\text{right\_hip}}) / 2, (y_{\text{left\_hip}} + y_{\text{right\_hip}}) / 2)$

## **Task 3 — AI Research**

### **1.Virtual Try-On Workflow**

Virtual try-on is a computer vision system that digitally places clothing onto a person's body.

**Input:** A person photo and a clothing image are taken as input.

**Human Parsing:** The body is segmented into regions like head, torso, arms and legs using a semantic segmentation model

**Pose Estimation:** MediaPipe detects 33 body landmarks and maps the skeleton structure onto the body.

**Clothing Warping:** The garment is deformed to fit the body shape using Thin Plate Spline transformation.

**Image Synthesis:** A GAN blends the warped cloth onto the person and generates the final try-on image.

### **2. Fashion Recommendation System**

It is like a smart shopping assistant that suggests clothes based on what you like and what fits your body.

**It studies you:** The AI looks at what clothes you liked, bought or searched for in the past and understands your style and preferences

**It compares you with others:** It finds other users who have similar taste as you and sees what they liked. If people similar to you liked a particular outfit then it recommends that to you too

**It checks your body:** Using pose detection it understands your body type and measurements and only recommends clothes that will actually fit you well

### **Research Papers**

- VITON — Virtual Try-On Network: <https://arxiv.org/abs/1711.08447>
- OpenPose — Body Landmark Detection: <https://arxiv.org/abs/1812.08008>
- Human Body Measurement Using Computer Vision: <https://arxiv.org/abs/2109.09913>
- Fashion Recommendation Using Deep Learning: <https://arxiv.org/abs/1907.05399>